



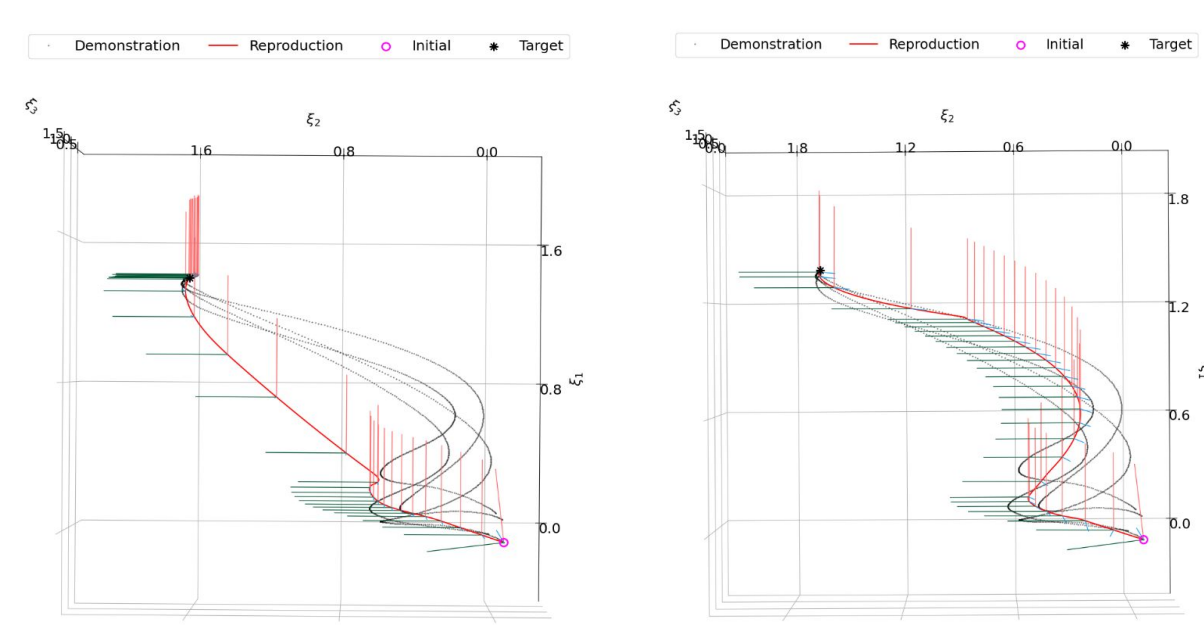
SE(3) LPV-DS for Quadrotor Systems

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MOTIVATION

- LPV-DS has mainly been used in over-actuated, holonomic systems (Kinematic Chains)
 - The capabilities of LPV-DS for other types of systems have been underrepresented
- Given the strong coupling between quadrotor motion and orientation, control policies should account for this interaction
- We aim to simulate quadrotor controls based on the generalized trajectories of the SE(3) LPV-DS model

Quadcopter Path Extrapolations
Euclidean vs SE(3) LPV-DS



RELEVANT BACKGROUND

LPV-DS (Euclidean)

- Models motion using a weighted combination of locally linear dynamical systems
- Global asymptotic stability is ensured by utilizing a Lyapunov function

Quaternion LPV -DS

- Extends LPV-DS to handle 3D orientations
- Uses Riemannian geometry on the unit quaternion manifold

SE(3) LPV-DS

- Integrates LPV-DS for positions and Quaternion-DS for orientations into a single dynamical system
- Fits a 6-DoF GMM to the demonstrations to learn full-pose, coupled motion policies

APPROACH

Generate demonstrations
using PID controller in
safe-control-gym



Train LPV-DS on
demonstrations to
generalize trajectories across
the state space



At runtime, use the DS model
to generate target velocities
based on position
(ignoring quaternion input)



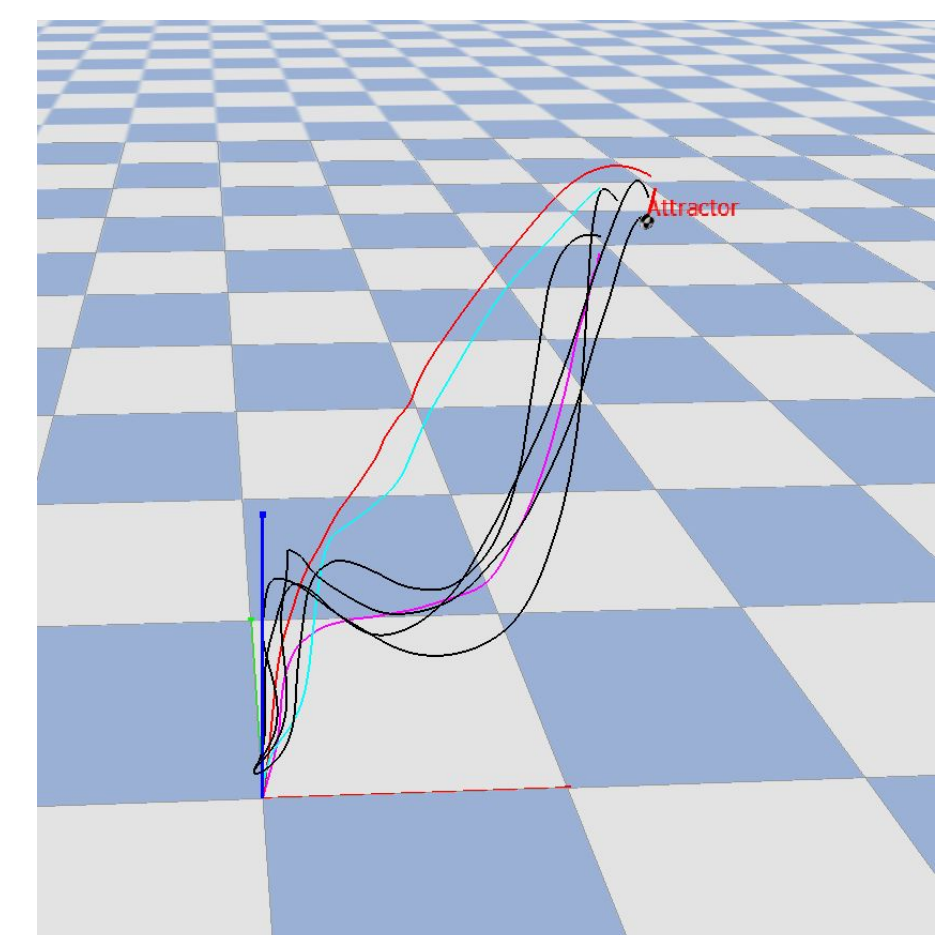
Use PID controller to
generate low-level control
for each rotor given target
velocities and quaternions

TOOLS

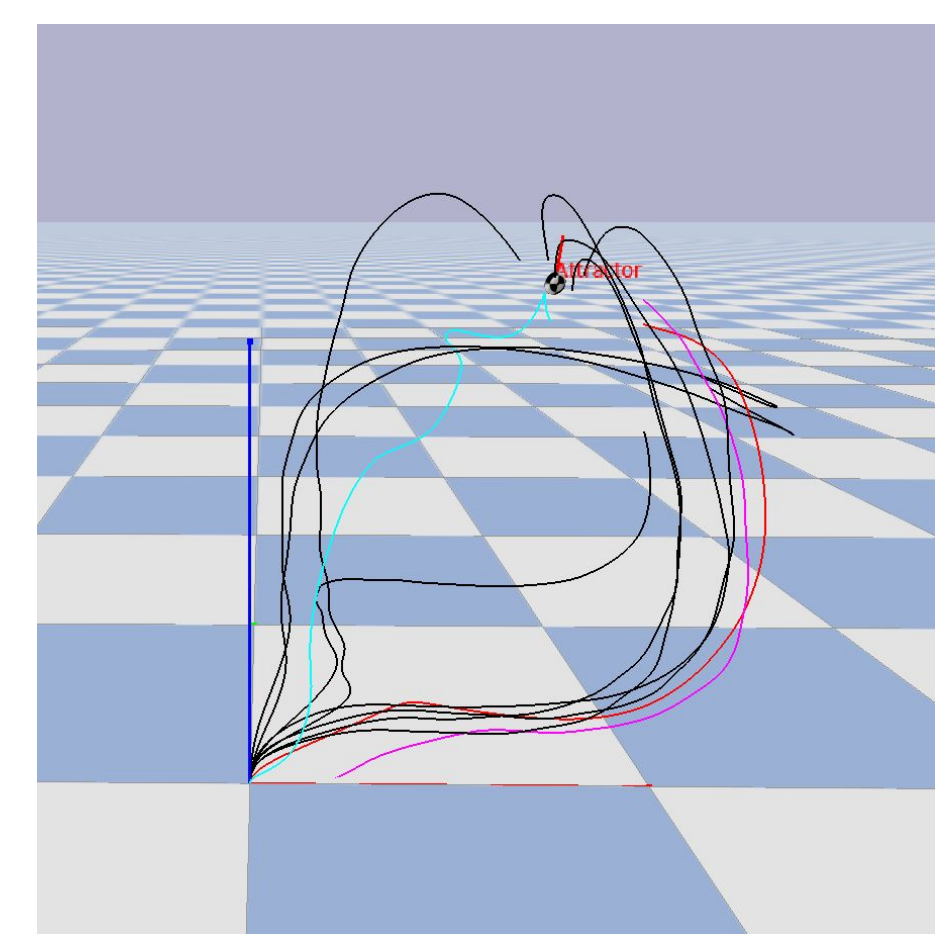
- We used the LPV-DS implementations provided by the Figueroa Lab and outlined in the paper *“SE(3) Linear Parameter Varying Dynamical Systems for Globally Asymptotically Stable End-Effector Control” (Sun & Figueroa)*
- For testing our implementation, we utilized **safe-control-gym**, a Pybullet-based environment for quadrotor controller simulation outlined in *“Safe-Control-Gym: A Unified Benchmark Suite for Safe Learning-Based Control and Reinforcement Learning in Robotics” (Yuan et al.)*

RESULTS

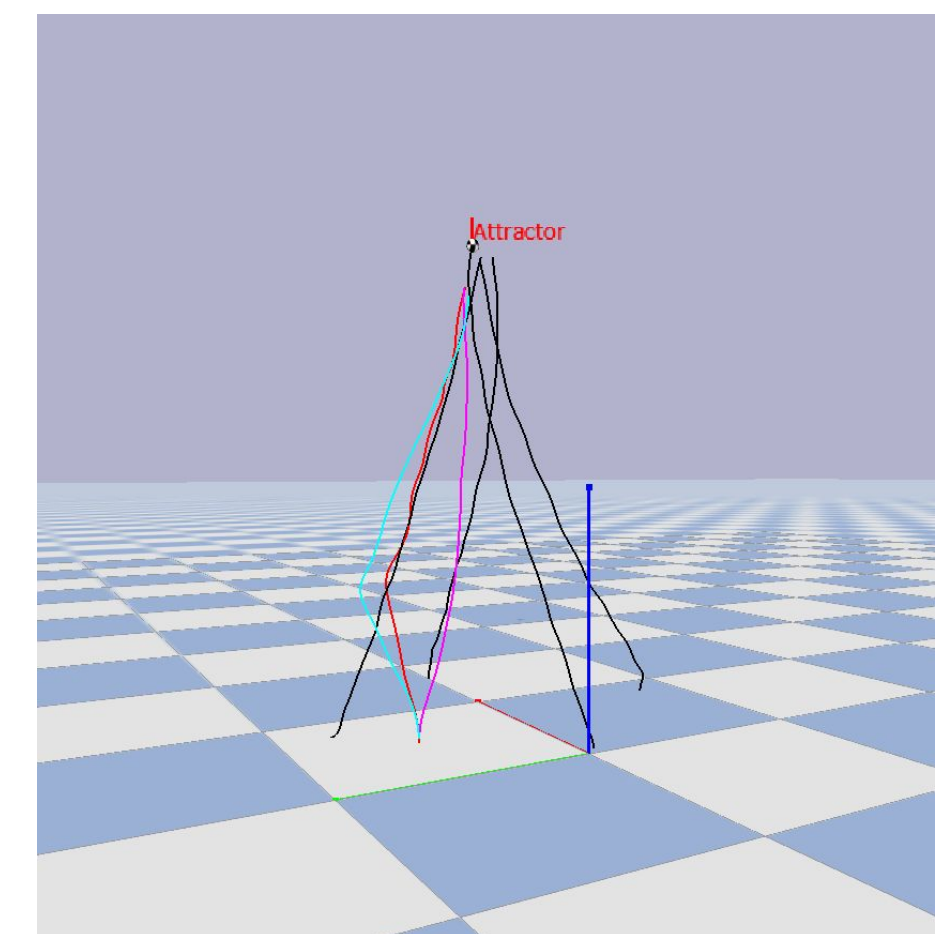
“S”-shaped



Box-shaped



Volcano-shaped



■ Demo ■ Coupled SE(3) LPV-DS
■ LPV-DS ■ DAMM Decoupled SE(3) LPV-DS

DISCUSSION AND FW

- Out of simulation, we can see that the SE(3) system qualitatively outperforms the Euclidean version
- In simulation, we can see that the decoupled SE(3) system performs better than the full SE(3)
- We suspect we can get greater performance for coupled system by incorporating a DAMM into the SE(3) workflow